Cohort-Anchored Infant EEG State-Space Geometry Tracks Age in Early Infancy

Child-clustered coordinate trajectories, with within-child local-dynamics robustness checks

Article type: Research Article

Short title: Cohort-anchored infant EEG geometry

Keywords: infant EEG, longitudinal development, state-space geometry, cohort anchor, GEE, exploratory analysis

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Abstract We examined age-associated change in EEG-derived state-space coordinates in 70 mixed rest/reaching recordings from 22 typically developing infants (38–203 days; 20 with repeated visits) in an independent Figshare cohort, after masking artifact-flagged (BAD_-interval) segments before epoching (5,265 of 6,929 candidate epochs retained). A frozen cohort-level IQR anchor placed every recording in a common 3D/9D coordinate chart. Child-clustered Gaussian GEE models with a cubic age spline showed age association at all 12 cohort-coordinate endpoints after FDR correction, driven by a within-child rather than between-child effect (within-child $p < 0.011$ at every endpoint; no between-child term FDR-significant). A more conservative child-cluster permutation check supported only 6 of 12 endpoints after its own FDR correction; the mobility and entropy-related coordinates were least robust, the latter also failing a direct-age sensitivity check. Two independent local-dynamics checks – within-child MNJ/reachability, and a cohort-pooled, leave-one-child-out reachability estimate – showed no supported, or even nominal, age effect; a nominal local-dynamics hint present before artifact masking disappeared afterward, consistent with a motion-artifact origin. These results show a within-child coordinate association that is robust for roughly half, but not all, endpoints under conservative testing, without an established corresponding change in local flow geometry.

Author Summary

We tracked EEG-derived state-space coordinates across infant recordings using a shared, cohort-level coordinate chart, after masking motion/artifact-contaminated segments. Coordinates changed with age within individual children, not just between them, but a stricter permutation-based check supported only half of the endpoints once corrected for multiple testing, flagging two (mobility, entropy-related) as fragile. Two checks of local flow-geometry summaries – within-child and cohort-pooled – found no supported change; a small hint seen before artifact masking disappeared once artifact-contaminated segments were removed.

1. Introduction

Longitudinal infant EEG can characterize developmental change, but irregular recording schedules make child-level inference and calibration especially important. We analyse an independent Figshare infant EEG cohort (Xiao & Smith, 2017) using exported EEG-derived state-space summaries. “State-space geometry” here denotes the geometry of a versioned feature chart, not a directly observed neural state or latent neural manifold.

Core question

Do common-scale, cohort-anchored MNPS coordinates show age-related change across this cohort, and does that change replicate as a within-child local-dynamics (MNJ/reachability) effect?

2. Theory

Let x_t denote the three-dimensional EEG-derived MNPS chart and $f(x,t)$ its estimated local flow. The MNJ is the local linearisation

$$J(x,t) = \partial \frac{f}{\partial} x. \tag{1}$$

We summarize its magnitude, trace, rotation, and spectral radius, and derive short-horizon reachability summaries from the associated local linear model. These operational statistics describe the exported chart and do not uniquely identify a biological mechanism.

3. Methods

3.1. Dataset and export contract

The Figshare source is a longitudinal typical-development dataset (Xiao & Smith, 2017). The analysed export contains 70 H5 recordings from 22 children, with visits 1–6 and ages 38–203 days. Sixty-nine geometry contracts were `ok` and one was `adjusted` under the standard invalidity policy; all were retained and all 70 visits provided 3D and 9D MNJs. The upstream export used 256-Hz EEG, 0.3–30-Hz filtering, 8-s windows with 4-s steps. This analysis uses an artifact-masked re-export: the upstream ingest pipeline's BAD_-interval annotations (715 artifact intervals, 2,143.7 s total, from the publicly released BAD_ files) are masked at the sample level before epoching, so any 8-s epoch overlapping a flagged interval is dropped rather than bridged; this reduced the usable epoch pool from 6,929 to 5,265 epochs (median 78.5 per session, range 20–109), down from a prior median of 98 (range 30–148) in the pre-masking export. The frozen subject-balanced IQR anchor `figshare_infant_eeg_global_cohort_all_visits_iqr_v2_1` was refit on this artifact-masked export (anchor hash `45bbfa21...c301a89e`) and provides both subject- and cohort-anchored layers. Rest/reaching condition labels are not yet available upstream; all recordings are analysed as mixed rest/reaching sessions, and BAD_ annotations are used only for artifact QC, not condition labelling.

3.2. Analysis design

The visit, not the epoch, was the inferential unit. The primary analysis is the cohort-anchored coordinate model: a Gaussian GEE with exchangeable child clustering and a 3-df cubic age spline fit to 3D and 9D cohort-coordinate medians, with FDR applied across the 12-coordinate family (Benjamini & Hochberg, 1995). Two robustness checks were added to this primary model: (1) a within/between-child decomposition that replaces the raw age spline with a linear child-mean ("between") age term plus a cubic spline of centered ("within") age, testing each component separately; and (2) a child-cluster row-permutation check of the omnibus Wald statistic (age values permuted across all visit-rows, the identical GEE refit 500 times per endpoint), which does not rely on asymptotic sandwich-covariance theory and is therefore a useful check given only 22 clusters.

As a robustness complement to the coordinate result, two independent local-dynamics analyses assessed whether the same age association is also present as a change in local flow geometry. The first is a within-child analysis of each session's own MNJ Frobenius norm, rotation norm, trace, spectral radius, and reachability log-volume, effective dimension, anisotropy, directional entropy, and rotational proxy (each session's reachability estimated only from that session's own epochs); it removes each child's mean age and endpoint value, then uses child-cluster bootstrap intervals and child-sign-permutation p-values. Twenty children (68 endpoint-complete visits) contributed to this within-child robustness check; full results, including a direct-age-only sensitivity subset, are reported in Supplement S2. The second is a cohort-pooled reachability analysis: at each visit's own cohort-anchored 3D coordinate, a local linear flow model is fit from the k=12 nearest epoch-to-epoch transitions drawn from every **other** child's epochs (leave-one-child-out), using the same ridge-regression and reachability-propagation machinery as the within-session estimate; the resulting per-visit reachability summaries are tested with the identical child-cluster estimator used for the within-child family.

3.3. Boundaries and robustness

Cohort-anchored coordinate positions are interpretable only under this export's frozen anchor and are not pooled numerically with APPLESEED (Puglia et al., 2022). Subject-anchored coordinates remain descriptive for within-child use. Cohort-coordinate robustness restricts the GEE analysis to 40 spreadsheet-direct age records; within-child robustness uses 36 eligible visits from 18 children. Bayley fields are retained for future work but are not analysed as outcomes.

4. Results

4.1. R1: Coverage and QC

The cohort had 70 visits from 22 children; 20 had repeated recordings. Age coverage was uneven and visit number was not a fixed developmental age stratum. All visits had resolved 9D MNJs; one was retained as an `adjusted` geometry contract after artifact masking (down from two in the pre-masking export).

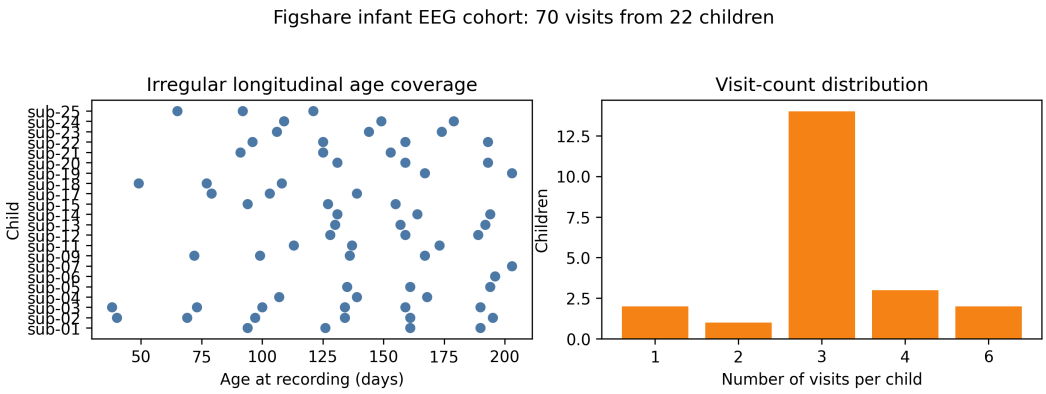


Figure 1: Age coverage and visit density. The irregular schedule motivates within-child age-days rather than visit-number inference.

4.2. R2: Cohort-anchored age trajectories (primary)

All 12 cohort-coordinate endpoints showed an age association in child-clustered spline GEE models after FDR correction. In 3D, deviation was the strongest coordinate result (Wald $\chi^2 = 54.72$, raw p=7.86 × 10⁻12, FDR p=4.72 × 10⁻11); entropy-related coordinate was more modest ($\chi^2 = 15.09$, FDR p=2.09 × 10⁻3); mobility, the weakest 3D result, improved after artifact masking relative to the pre-masking export ($\chi^2 = 11.28$, raw p=0.0103, FDR p=0.0112). The strongest result overall was 9D m_o ($\chi^2 = 155.37$, FDR p=2.19 × 10⁻32); the weakest was 9D d_s ($\chi^2 = 7.89$, raw and FDR p=0.048). These are common-chart position associations, not direct local-flow or neural-state effects.

The within/between decomposition (Methods) shows this is a genuine within-child effect: at every one of the 12 endpoints the within-child spline term is significant (all raw p < 0.011; strongest: 9D m_o, $\chi^2 =$ 337.4, p=7.94 × 10⁻73; weakest: 9D d_s, $\chi^2 = 11.25$, p=0.0105), while no endpoint's between-child linear term is FDR-significant (raw p range 0.04-0.98). The child-cluster row-permutation check is a substantially more conservative, finite-sample-robust test of the same omnibus statistic: after BH-FDR correction of the 12 permutation p-values, 6 of 12 endpoints remained supported – 9D m_o (permutation FDR p=0.024), and 3D deviation, 9D m_e, 9D d_n, and 9D e_e/e_m (permutation FDR p=0.040 each). Six endpoints did not survive: 3D mobility and 3D entropy-related coordinate (permutation FDR p=0.152 each), 9D m_a and 9D e_s (p=0.075 each), 9D d_l (p=0.062), and 9D d_s (p=0.222). We therefore report the cohort-coordinate result at two confidence levels: the asymptotic GEE supports all 12 endpoints and a within/between decomposition confirms the effect is within-child rather than cross-sectional at all 12, but a conservative FDR-corrected cluster-permutation check supports only half (6 of 12). Full per-endpoint decomposition and permutation tables (including the FDR-corrected permutation p-values) are in Supplement S2.

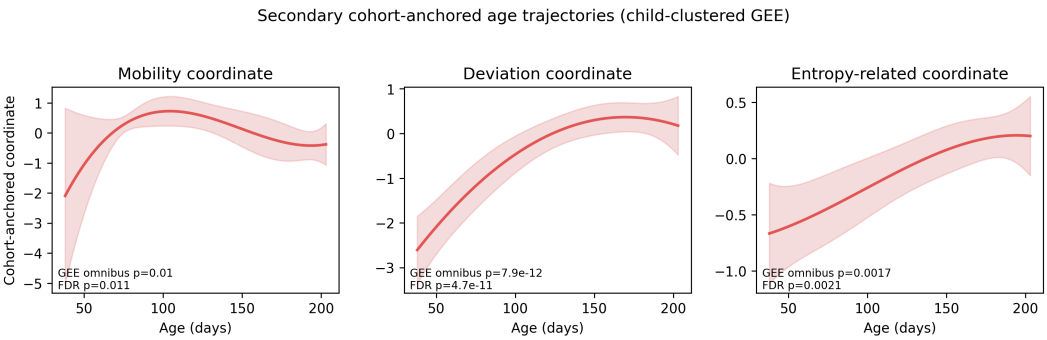


Figure 2: Cohort-anchored 3D coordinate trajectories from child-clustered spline GEE, the primary result of this study. Curves describe common-chart coordinate position, not directly observed neural states.

4.3. R3: Age-source sensitivity of the cohort-coordinate result

In the 40-record spreadsheet-direct age sensitivity, 11 of 12 cohort-coordinate models remained FDR-supported; only the 3D entropy-related coordinate was not (raw/FDR p=0.070). The 3D mobility coordinate was more strongly supported under this sensitivity (FDR p=6.62 × 10⁻4) despite failing R2′s cluster-permutation check, whereas the entropy-related coordinate now fails **both** the direct-age sensitivity and the cluster-permutation check, making it the single least robust endpoint in this analysis. The two checks probe different aspects of

robustness (age-source dependence vs. finite-sample cluster inference) and need not agree, but an endpoint failing both warrants particular caution.

4.4. R4: Within-child local-dynamics robustness check

As a robustness complement to R2-R3, we asked whether the cohort-level age association also appears as a within-child local-dynamics effect estimated from each session's own epochs. None of the nine primary 3D MNJ/reachability endpoints showed a nominal within-child age association, and none was even close to nominal significance after artifact masking (smallest raw p, MNJ trace: standardized slope −0.22, raw p=0.054, FDR sensitivity p=0.484); a secondary 9D MNJ sensitivity family and a direct-age-only within-child subset were also tested, with no FDR-supported endpoint, and no nominally significant endpoint at all, in either (smallest raw p=0.076 and 0.208, respectively). Notably, two 3D endpoints that were nominally significant (uncorrected) in the direct-age subset of the pre-masking export (`frobenius_norm_median` and `rotation_norm_median`, raw p=0.043-0.050) are no longer nominally significant after artifact masking (raw p=0.277 and 0.208); this pattern is consistent with those earlier nominal hits having been driven by residual motion artifact rather than a developmental effect, though this cannot be confirmed directly and the corrected/masked comparison is descriptive, not a formal before/after test. Full results, figures, and the direct-age within-child sensitivity numbers are in Supplement S2. This null is compatible with, and does not contradict, the R2 cohort-level result: developmental movement across cohort-chart position need not imply a detectable within-child change in the local Jacobian or short-horizon reachability summaries at this sample size.

4.5. R5: Cohort-pooled reachability robustness check

As a second, independent local-dynamics test, we estimated reachability at each visit's own cohort-coordinate position from a local flow model fit exclusively to **other** children's nearby epoch-to-epoch transitions (leave-one-child-out; median 8 distinct contributing children per visit, from a pool of 5,195 transitions). None of the five cohort-pooled reachability endpoints was FDR-supported, and none was nominally significant (smallest raw p, anisotropy: standardized slope 0.27, raw p=0.478); the direct-age-only subset showed a single nominal (uncorrected) hit for the rotational-proxy endpoint (standardized slope −0.47, raw p=0.040, FDR sensitivity p=0.200), not FDR-supported. The pooled neighbor radius was stable across age terciles (median 0.35, 0.29, and 0.33 in the youngest, middle, and oldest thirds of the sample), so these nulls are not attributable to markedly uneven cohort density at different ages.

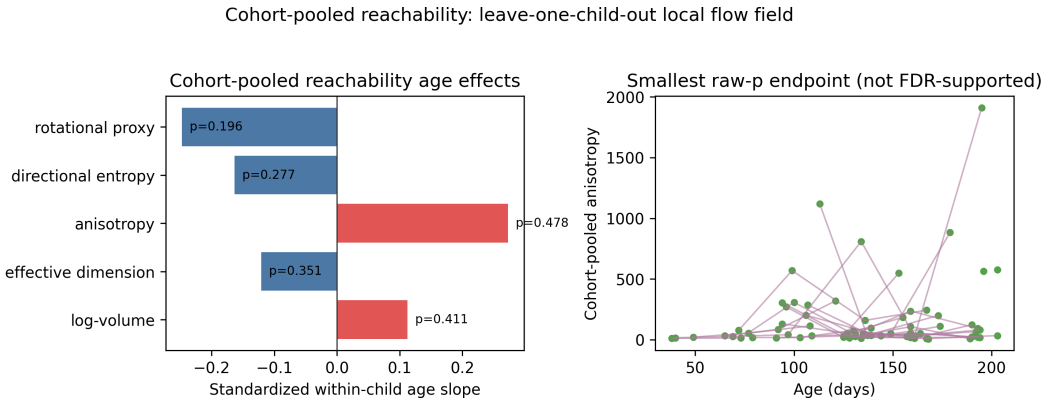


Figure 3: Cohort-pooled reachability age effects (leave-one-child-out local flow field) and the family's own smallest raw-p endpoint (anisotropy), not FDR-supported.

5. Discussion

The primary result of this reanalysis is that cohort-anchored coordinate positions show age association in a common EEG-derived chart across 70 visits from 22 children, and a within/between-child decomposition shows the effect is driven by within-child movement rather than between-child differences at all 12 endpoints. That said, the strength of evidence depends materially on which robustness check is applied: the asymptotic FDR-corrected GEE supports all 12 endpoints, but a child-cluster row-permutation check – designed specifically because 22 clusters is a known regime for anti-conservative sandwich-covariance inference – supports only 6 of 12 endpoints once its own p-values are FDR-corrected. This is exactly the small-cluster fragility that motivated adding the check, and it substantially tempers the "all 12 endpoints" framing of the asymptotic result: readers should treat 9D m_o as the single most robust result, 3D deviation and 9D $m_e/d_n/e_e/e_m$ as moderately robust, and the remaining six endpoints (most notably 3D mobility and 3D entropy-related coordinate, which also fails the direct-age sensitivity check) as supported only by the asymptotic test.

This reanalysis was itself motivated by an artifact-QC improvement: masking EEG segments flagged as motion/artifact-contaminated by the upstream BAD_-interval pipeline removed roughly a quarter of the epoch pool (6,929 to 5,265 epochs) and slightly **strengthened** the primary cohort-coordinate result (e.g. 3D mobility's asymptotic FDR p improved from 0.034 to 0.011), while a previously-noted nominal signal in the within-child local-dynamics direct-age sensitivity subset (two 3D endpoints at raw p=0.043-0.050) disappeared entirely (both now raw $p > 0.2$). Both changes point the same direction: the coordinate-level developmental association is not an artifact of residual movement contamination, and the earlier nominal local-dynamics hint plausibly was.

Two independent local-dynamics checks – one estimated purely from each session's own epochs (within-child), one estimated from a leave-one-child-out cohort-pooled flow field at each visit's own coordinate – did not detect a corresponding age effect at this sample size, and neither showed even a nominal signal after artifact masking. Because the cohort-coordinate model pools within- and between-child age information while both local-dynamics checks isolate a different, more local estimand, these are compatible but distinct results, not a contradiction: cohort-chart position moves with age, in a way that is robustly attributable to within-child change, without a correspondingly detectable change in either session-level or cohort-pooled local flow geometry.

The main internal contribution is a contract-aware pipeline: it retains session-level inference, freezes a subject-balanced common anchor, masks artifact-flagged segments before epoching, separates chart-position from local-dynamics claims, decomposes within- from between-child age effects, adds a permutation-based robustness check to the asymptotic GEE test, and tests local dynamics both within-child and cohort-pooled. A future age-balanced cohort with an independently fitted or cross-fitted anchor should test whether the R2 coordinate trajectories, and the R2 within/between decomposition, replicate.

6. Limitations

The source combines mixed rest/reaching EEG without event-level labels (BAD_-interval annotations are used only for artifact QC, not condition labelling), lacks stage labels, and has uneven follow-up. Thirty age values are derived from first-visit age and recording date. The cohort anchor was fit using all usable ages, so it is suitable for descriptive common-scale association but is not a held-out normative reference. The spline is exploratory. Only 6 of the 12 cohort-coordinate endpoints survive a conservative, FDR-corrected child-cluster permutation check of the omnibus test, even though all 12 pass the asymptotic FDR-corrected GEE and the within/between decomposition; the 3D mobility and 3D entropy-related coordinates are the least robust (the latter also fails the direct-age sensitivity check) and should be treated as fragile pending replication, and the remaining endpoints supported only by the asymptotic test should be read with corresponding caution. The cohort-pooled reachability estimate (R5) uses a single local fit per visit rather than a within-session tube of estimates, so it is not a strictly larger-sample version of the within-child check (R4) but a distinct estimand with its own assumptions, most notably that leave-one-child-out neighbor density is roughly comparable across ages (checked descriptively, Results R5). Both local-dynamics robustness checks (R4, R5, Supplement S2) are limited by modest child count and irregular sampling and should not be read as evidence of developmental invariance in local dynamics. The comparison between this artifact-masked export and the earlier pre-masking export is descriptive, not a formal statistical test of the effect of artifact masking, since it involves two different, non-independent versions of the same underlying recordings rather than a controlled ablation.

7. Conclusion

In this independent infant cohort, cohort-anchored MNPS coordinate positions changed with age across children, and this effect is attributable to within-child movement rather than between-child differences at every endpoint tested. However, only 6 of 12 endpoints remain supported once a conservative, FDR-corrected child-cluster permutation check is applied, and the 3D mobility and entropy-related coordinates are fragile under at least one robustness check each. Two independent local-dynamics robustness checks – within-child and cohort-pooled – showed no supported, or even nominal, effect at this sample size after artifact masking, and a previously-noted nominal local-dynamics hint disappeared once artifact-contaminated segments were removed. This is evidence of a within-child developmental association in a common coordinate chart for a subset of endpoints, not a robustly established effect across the full 12-endpoint family, and not yet a confirmed local-dynamics or neural mechanism.

8. Acknowledgements

We acknowledge Xiao, Smith, and collaborators for the Figshare dataset and the upstream metadata mapping.

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Author Contributions

RL: Conceptualization, Methodology, Data Curation, Analysis, Writing, and Editing.

Funding Information

No dedicated funding supported this exploratory reanalysis.

Data Availability Statement

The source data are available through Figshare (Xiao and Smith, 2017). Analysis wrappers and derived result tables are in this repository under `articles/infant_eeg/`.

Competing Interests

The author declares no competing interests.